

Using Mean Orbit Period in Mars Reconnaissance Orbiter Maneuver Design

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Mars Reconnaissance Orbiter (MRO) has provided communication relays for a number of Mars spacecraft. In 2016 MRO is expected to support a relay for NASA's Interior Exploration using Seismic Investigations, Geodesy and Heat Transport (InSight) spacecraft. In addition, support may be needed by another mission, ESA's ExoMars EDL Demonstrator Module's (EDM), only 21 days after the InSight coverage. The close proximity of these two events presents a unique challenge to a conventional orbit synchronization maneuver where one deterministic maneuver is executed prior to each relay. Since the two events are close together and the difference in required phasing between InSight and EDM may be up to half an orbit (yielding a large execution error), the downtrack timing error can increase rapidly at the EDM encounter. Thus, a new maneuver strategy that does not require a deterministic maneuver in-between the two events (with only a small statistical cleanup) is proposed in the paper. This proposed strategy rests heavily on the stability of the mean orbital period. The ability to search and set the specified mean period is fundamental in the proposed maneuver design as well as in understanding the scope of the problem. The proposed strategy is explained and its result is used to understand and solve the problem in the flight operations environment.

Nomenclature

<i>AMD</i>	=	Angular momentum desaturation
<i>APSM</i>	=	Absolute Period Synchronization Maneuver
<i>DCO</i>	=	Data Cutoff
<i>EDL</i>	=	Entry, Descent and Landing
<i>EDM</i>	=	EDL Demonstrator Module
<i>InSight</i>	=	Interior Exploration using Seismic Investigations, Geodesy and Heat Transport
<i>LMST</i>	=	Local Mean Solar Time
<i>Ls</i>	=	Solar Longitude
<i>MRO</i>	=	Mars Reconnaissance Orbiter
<i>OD</i>	=	Orbit Determination
<i>PSM</i>	=	Period Synchronization Maneuver
<i>RPSM</i>	=	Relative Period Synchronization Maneuver
<i>RSS</i>	=	Root Sum Square
ΔV	=	Delta-V, change in velocity
ΔP	=	Delta-Period, change in mean period

I. Introduction

THE Mars Reconnaissance Orbiter (MRO) spacecraft is expected to provide telecommunication relay support for NASA's Interior Exploration using Seismic Investigations, Geodesy and Heat Transport (InSight) spacecraft during the Entry, Descent and Landing (EDL) sequence in September 2016. In addition, it may support ESA's ExoMars EDL Demonstrator Module's (EDM) EDL phase or the first overflight of the EDM landing site in October

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2016^{*}. In the past, MRO has performed relay telecommunication support for other vehicles arriving at Mars: Mars Phoenix Lander in May 2008¹ and Mars Science Laboratory in August 2012.² Each of these support campaigns required a series of phasing maneuvers to synchronize the MRO orbit with a set of desired orbit targets at a specified target epoch prior to the incoming vehicle's atmospheric entry. Unlike these past support efforts, MRO will be performing phasing maneuvers to achieve two events separated by only 21 days: InSight EDL on September 28, 2016 and the first overflight of the EDM landing site on October 19, 2016. Since the InSight coverage is required to last for at least a week after the InSight EDL, there remains at most 14 days between the phasing maneuver for EDM and the EDM EDL/first overflight. If the time interval between the InSight and EDM events gets shortened or the InSight coverage duration gets lengthened, the time interval between the phasing maneuver for EDM and the actual EDM EDL may get shortened. And the shorter this interval is, the higher the ΔV of the phasing maneuver will be; in addition, the timing error may increase due to the maneuver uncertainty from the higher ΔV . Thus, a maneuver strategy that requires no deterministic maneuver between the two close proximity events was investigated.[†]

While planning MRO's navigation support of both InSight and EDM EDLs, a maneuver strategy with a sequence of two maneuvers was proposed. The first is an absolute-phase synchronizing maneuver (APSM) and the second is a relative-phase synchronizing maneuver (RPSM). These two maneuvers are designed conceptually in reverse order; RPSM is designed before APSM. To construct these two maneuvers, a maneuver search method was implemented that finds a ΔV to set a specified mean period of the MRO orbit. This paper presents the details of this maneuver strategy along with the implementation and results.

II. Background

This section provides some pertinent background information to help the reader better understand the material presented in the following section. It includes some details on the Mars atmospheric drag, MRO downtrack timing error propagation, MRO mean orbital period, and the reference trajectories used for the analyses.

A. Mars Atmospheric Drag

MRO has been orbiting continuously at Mars since 2006. As a result, the MRO Navigation team now has several years of reconstructed atmospheric density information pertinent to predicted atmospheric modeling. For atmospheric modeling, MRO Navigation uses the NASA Marshall Space Flight Center's Mars Global Reference Atmospheric Model (Mars-GRAM) 2005 (Mapping Year 0). The predicted monthly solar flux values have been infrequently updated in the MRO Navigation's Mars-GRAM model setup in order to preserve and observe density trending. When 2-way Doppler tracking data is available (provided by the Deep Space Network), a scale factor of the estimated atmospheric drag ΔV per orbit relative to the predicted atmospheric drag ΔV is computed. The MRO orbit atmospheric drag ΔV (in mm/sec/orbit) for mission-to-date is plotted against solar longitude (Ls)[‡] in Figure 1. By plotting against Ls the density trends over years of operation at Mars are clearly seen.

The MRO orbit typically experiences "high" and "low" density seasons, which refer to seasons of higher and lower atmospheric drag ΔV imparted on the MRO spacecraft. Orbit phasing activities for Phoenix were conducted during the Primary Science Phase, and activities for MSL were conducted during Extended Mission 1 (EM1). Phoenix landing occurred at Ls = 76.3 degrees and MSL landing occurred at Ls = 150.6 degrees, both during the low-density season. However, InSight EDL will occur at 231.2 degrees and EDM will occur at 244.6 degrees, which are at the start of the high-density season. Anticipated mean atmospheric drag ΔV per orbit will be more than double the anticipated values used for long-term modeling in the design of the Phoenix and MSL orbit phasing maneuvers. Thus, when the MRO downtrack timing error for EDL is estimated, the value of 0.5 mm/s per orbit would be more appropriate near the EDL coverages, whereas a slightly lower value is proper for a few months prior to the events.

^{*} It is yet to be confirmed if the EDM support is at the EDL phase or at the first overflight. Nevertheless, the problem in question remains essentially the same.

[†] Note that these phasing maneuvers do not alter the node of the MRO orbit. It is assumed that the necessary correction to the node has been already initiated by a plane-change maneuver to reach the requirement established by InSight, 2:30 P.M. Mean Local Solar Time (LMST).

[‡] Ls conveys a Martian season and is the angle between Mars₀-Sun-Mars where Mars₀ is Mars at Ls=0° (Spring); thus, Ls=90° (Summer), Ls=180° (Fall), and Ls=270° (Winter). The seasons are with respect to the Northern Hemisphere.

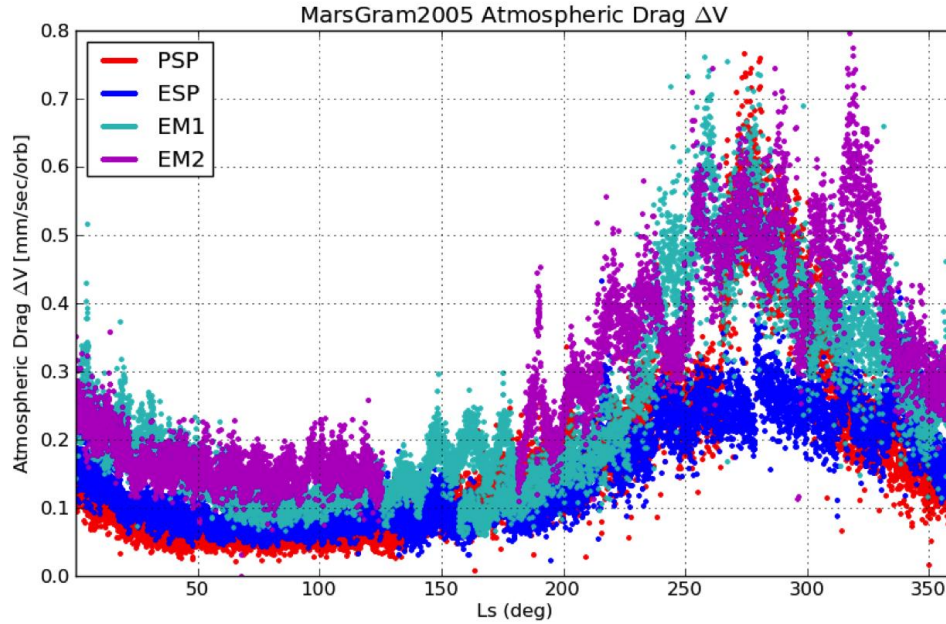


Figure 1. Navigation Reconstructed Atmospheric Drag ΔV per Orbit vs. Solar Longitude (Ls). *InSight & EDM EDLs occur at the start of a high atmospheric drag season where the drag ΔV is nearly 0.5 mm/s per orbit. Different colors denote different phases of the MRO mission (PSP= Primary Science Phase; ESP=Extended Science Phase; EMx=Extended Mission x); values are shown thru May 2014.*

B. MRO Downtrack Timing Uncertainty

Atmospheric drag ΔV , as well as components of angular momentum desaturation (AMD) events, maneuver execution error, and orbit determination (OD) error, impacts the orbit downtrack event timing. Refer to Ref. 2 for a more detailed introduction to these components and how they are computed and propagated to estimate the MRO downtrack timing uncertainty. The ΔV is mapped to orbit period error using basic orbit dynamics. For long-term downtrack orbit uncertainty modeling, MRO Navigation models atmospheric drag ΔV uncertainty in two ways: as a bias (constant) term and as a white-noise (orbit-to-orbit variation) term. MRO Navigation uses a figure of 35% variation (1σ) for orbit-to-orbit atmospheric drag ΔV variations and a figure of 10% (1σ) for a constant atmospheric drag ΔV bias. These figures were re-assessed during MSL support using 3 years of low-density season reconstructed MRO trajectory data; values will be re-assessed for InSight and EDM support using density season data appropriate for landing conditions later in the Martian year.

It is the execution error from a large ΔV correcting the EDM phase error, which arises due to the proximity of these two events, in combination with the high atmospheric drag ΔV during the InSight and EDM coverage, that causes the downtrack timing uncertainty to increase very quickly. In fact, the rapid increase in the timing uncertainty is mostly independent of the maneuver time. That is, if we delay the maneuver time to reduce the downtrack timing uncertainty, the maneuver execution error will grow due to the increasing maneuver ΔV . On the other hand, if we execute the maneuver early to minimize the maneuver ΔV magnitude, the length of the downtrack timing uncertainty grows. As a result, no matter when the phasing maneuver is performed in-between the EDLs, the resulting timing uncertainty at the EDM EDL remains nearly constant. For example, if we assume that we have to correct up to a $\frac{1}{2}$ orbit period within the 21-day separation between the two events, a constant timing error of about 70 seconds is expected at the EDM coverage no matter when the maneuver is executed within the 21-day period. Similarly, correct up to $\frac{3}{4}$ orbit period, we expect a constant downtrack timing error of about 100 seconds no matter when the maneuver is executed with the 21-day period. Refer to Figure 2 below for the propagation of the downtrack timing error uncertainty for correcting $\frac{3}{4}$ orbit period in 10 days and 14 days, respectively. Figure 3 shows that the timing uncertainty remains nearly constant. This is a primary motivation for seeking an alternate strategy that requires no deterministic maneuver between the two EDLs.

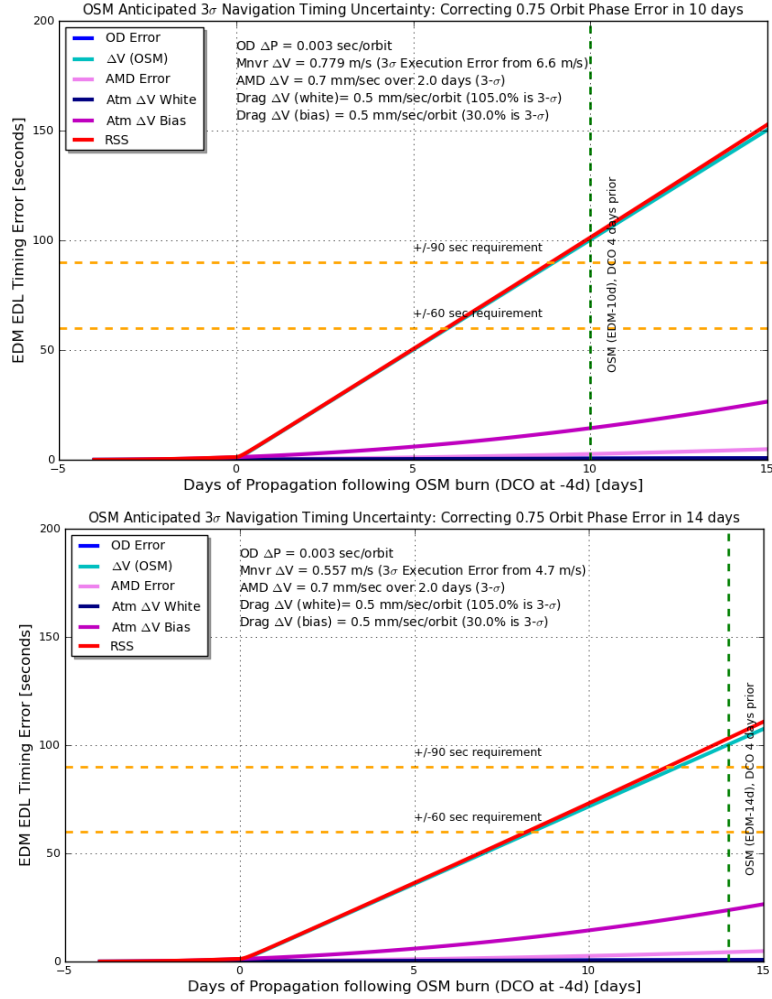


Figure 2. Downtrack Timing Uncertainty Propagation for Correcting $\frac{3}{4}$ Orbit Period in 10 & 14 days. *The execution error from the maximum ΔV dominates the uncertainty. Note that the timing error at the end of propagation is nearly the same in both cases.*

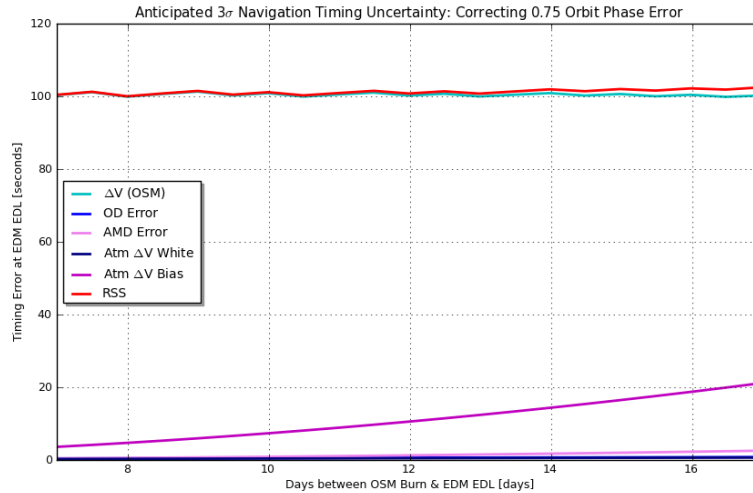


Figure 3. The Maximum Timing Uncertainty at the End of Propagation Correcting $\frac{3}{4}$ Orbit Phase. *Note that the downtrack timing uncertainty remains nearly constant at about 100 seconds no matter when the maneuver is performed within the 21 days.*

C. MRO Mean Orbital Period

The mean orbital period removes the periodic oscillations of the osculating orbital period arising from various perturbing forces. The osculating-to-mean-element conversion was performed with MRO flight operations software that converts the J_2 part of the gravity field³ as well as all the other gravity harmonics with the linearized solutions⁴. Figure 4 below shows the shorter term stability of the mean period within a dozen orbits. The mean period remains nearly constant where the osculating period varies aggressively, as a function of the true anomaly.

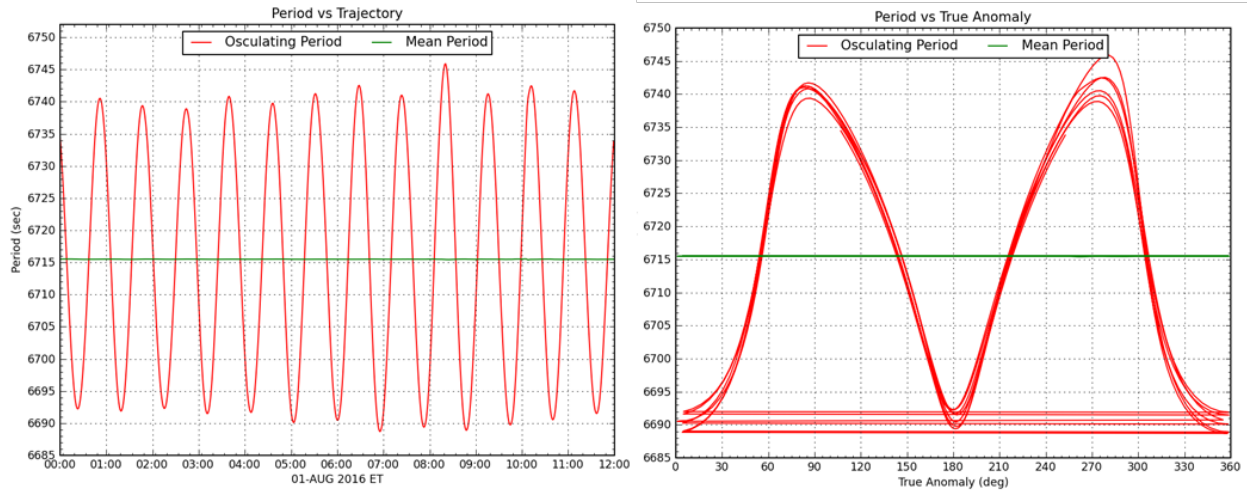


Figure 4. Short-Term Osculating Period vs Mean Period. *The left plot shows osculating period vs mean period along the timeline whereas the right plot shows it along the true anomaly.*

Figure 5 below shows the longer term stability of the mean period. It is nearly linear; however, there is a small slope of about 1.5 seconds in 3 months, mainly due to the atmospheric drag. Also note that the slope is slightly greater in the second half of the plot as MRO enters the high atmospheric drag season in September 2016.

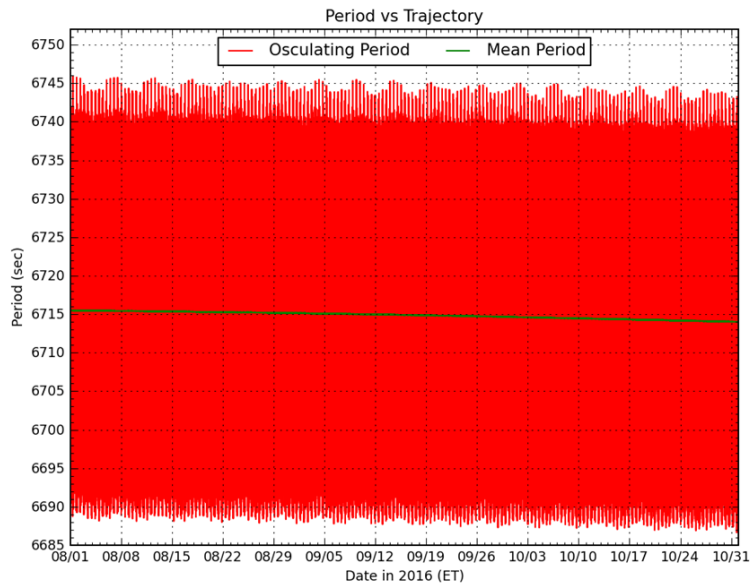


Figure 5. Long-Term Osculating Period vs Mean Period. *The plot represents the time period around the InSight and EDM EDLs. Note that the mean period is linear with a slight downward slope due to the atmospheric drag.*

* Another (newer) osculating-to-mean conversion software by Todd Ely (JPL) was briefly investigated and tried but not used for this analysis.

There does exist a tiny variation of the mean period on a scale not visible on the above plots. For this reason, the mean periods are computed over ± 2 hours at 60 points and averaged for further stability. Thus, to estimate the maneuver ΔV in correcting a phase Φ in N number of orbits simply requires the change in the mean period of Φ / N . A working conversion factor between ΔV and ΔP (change in mean period) for MRO is about 1 m/s of ΔV per 6 seconds of ΔP . The maneuver strategy proposed in this paper takes advantage of this stability and linearity of the mean period for the design and estimate of the maneuver ΔV in correcting the phase.

D. Reference Trajectories

The MRO reference trajectory used for the analyses in this paper was derived from the InSight Mission support reference trajectory. This reference trajectory includes about a 5.5 m/s plane change maneuver in July 2015 to transition MRO to a 2:30 PM Local Mean Solar Time (LMST) at the ascending node of the InSight EDL from its nominal 3:00 PM LMST.

In reality, it is likely that the final phasing of the MRO trajectory at the time of the InSight EDL will vary significantly from this reference trajectory due to the uncertainty of events between now and September 2016. In particular, MRO has already begun to implement a 26-minute phase delay to “hide” behind Mars for the Comet Siding Spring (CSS) encounter in October 2014, causing $\frac{1}{4}$ orbit bias from the current reference trajectory.

Thus, out of the InSight-support reference trajectory several derived trajectories were created to examine the effect of differently phased trajectories. They include the following:

- 1) Reference trajectory propagated with the atmospheric drag (*reference_atm*). Note that the reference trajectory did not model the atmospheric drag to simulate the ground track walk maintenance; in reality Orbit Trim Maneuvers are performed every 8 weeks as needed to maintain the ground track walk, which gets altered due to the atmospheric drag.
- 2) Reference trajectory phased to account for the amount shifted for CSS (*phase0*) with the atmospheric drag modeled in from April 2016, when we tentatively plan to start considering the problem realistically after the InSight launch. This is the most probable trajectory at the present time.
- 3) Other phased trajectories (*phase $\pm\Phi$*) with respect to the *phase0* trajectory where Φ represents the phase shift by 10, 20, and 30 minutes. These trajectories were created mainly to observe the effect of slightly differently phased trajectories on the proposed maneuver strategy.

III. Maneuver Strategy Investigated

In this section, the proposed maneuver strategy and its implementation are explained in more detail. The result of the strategy is explored to understand the scope of the problem. Finally, the robustness of the strategy is investigated via downtrack timing uncertainty propagation.

A. Strategy Illustrated

Stated simply, to do away with a phasing maneuver in-between the two EDLs, the MRO orbital period between the two events must be synchronized such that the time duration between the two events are exactly multiples of the orbit period. Or, put in another way, the duration between the two events divided by the number of orbits must coincide with the orbit period.

As already briefly introduced, the strategy proposes a sequence of two maneuvers. The first is an absolute phasing adjustment (APSM) and the second is a relative phasing adjustment (RPSM). The strategy is better understood conceptually when the second maneuver, RPSM, is designed before the first maneuver, APSM. To visualize the problem more succinctly, if we assume that both the InSight EDL and the EDM EDL occur at the same latitude, then, we will be able to synchronize the two EDL times by setting the mean period of the MRO orbit prior to the InSight EDL such that the integer multiples of the mean period equals the time interval between the two EDLs. To generalize the problem to the different EDL latitudes, we can simply take into account a fraction of the mean period, which amounts to the difference in the EDL latitudes. Thus, for any given time interval between the InSight EDL and the EDM EDL, there exists a mean period of the MRO orbit that best synchronizes the relative phasing between the two EDLs. Figure 6 below illustrates the results of the RPSM.

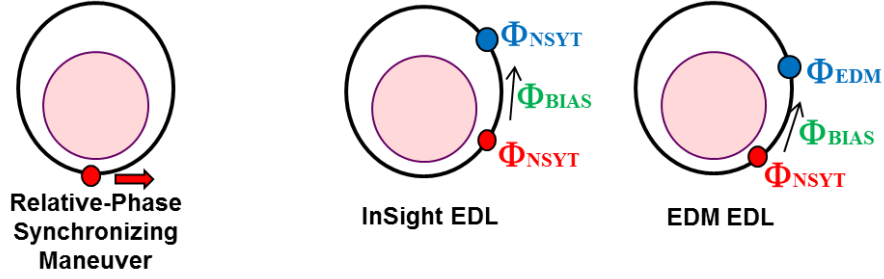


Figure 6. Relative-Phase Synchronizing Maneuver. The RPSM maneuver places MRO at the correct relative phasing (red); however, there still remains a phase bias (green) from the required phase (blue). The time proceeds from left to right (orbits not drawn to scale).

However, although RPSM synchronizes the relative phasing between the two EDLs, it cannot zero out the absolute phasing. In other words, the same phase bias remains at both EDLs. Thus, another maneuver must be used to remove this (absolute) phase bias. Since the new maneuver cannot alter the mean period set by RPSM, it must be placed prior to RPSM. This new APSM must remove the phase bias created by RPSM. The phase bias is removed at the time of RPSM. Figure 7 below illustrates the results of the APSM.

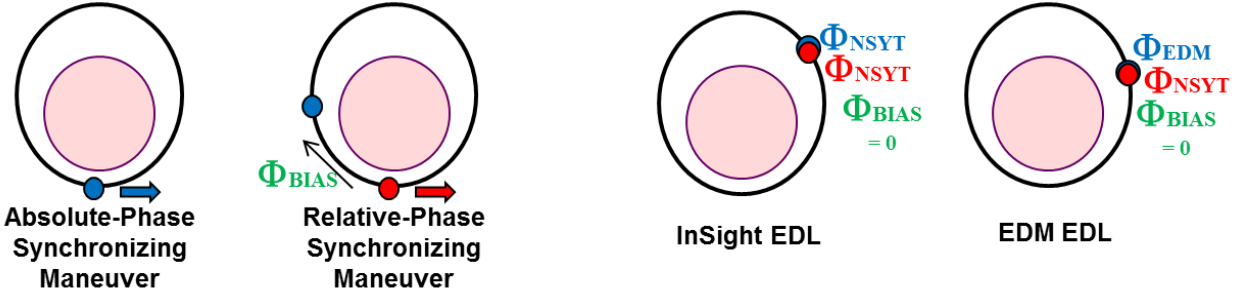


Figure 7. Absolute-Phase Synchronizing Maneuver. The APSM maneuver adds a negative phase bias at the time of RPSM so that, when RPSM is applied and propagated to the time of EDLs, the phase biases created originally by RPSM are removed at each EDL.

It is the usefulness of the RPSM to set the MRO mean period to a specified target value that makes this strategy work conceptually. Thus, although the addition of a new maneuver, APSM, prior to RPSM may alter the mean period leading up to the RPSM time, the mean period subsequent to RPSM should not be altered.

B. Implementation

The implementation proceeded in the following four steps:

- 1) Implement the mean-period-setting maneuver.
- 2) Implement the RPSM.
- 3) Implement the APSM.
- 4) Verify that the strategy actually works when all are pieced together.

The implementation of the mean-period-setting maneuver design was achieved by providing the mean period to the burn searching functionality in Monte.* The RPSM was implemented to successfully remove the “excessive” phase at EDM. The excessive phase refers to any amount of phase that is either above or under the multiples of orbital period between the two EDLs, whichever is closer. At this point, the RPSM synchronizes the mean period such that after the execution of RPSM it is guaranteed that the two EDL events will be synchronized under the assumption that the timing error is zero. (Of course, in reality the variation in the atmospheric drag and the maneuver execution error, etc. will bring the trajectory out of sync, which is why a cleanup maneuver is placed at a reasonable distance away from the maneuver itself as necessary.)

* Monte (Mission analysis, Operations, and Navigation Toolkit Environment) is the next-generation JPL Mission Design and Navigation software.

A more normal phasing maneuver, APSM, was also implemented using the mean-period-setting maneuver since it worked so well for the RPSM. However, when the two maneuvers (RPSM and APSM designed independently with RPSM before APSM) were executed together, there resulted a small random phase error at both events on the order of several seconds to up to about a minute. This is mainly attributed to the shift in the maneuver time (around an apsis) of RPSM when APSM is introduced before RPSM, which was designed with the original trajectory before the insertion of APSM. So, the original idea of simply independently designing two burns did not work precisely enough. Thus, an iterative method had to be introduced. This led to an investigation of a way to estimate the ΔP , given only the reference trajectory and computing the mean periods at a few different places in the reference trajectory, without actually propagating the trajectory. It turned out that the estimate (arithmetic using the mean period) worked so well that its error was within a few percent (or up to about 0.01 m/s when the burn is small) of the fully optimized solution with an iterative method. The iterative algorithm to solve for the fully optimized solution goes as follows: First, the original trajectory is propagated with the initial guess for APSM. Second, the trajectory with an estimated APSM is used to fully remove the excessive phase at EDM. The trajectory will then contain a small phase bias at both InSight and EDM EDLs. Third, APSM again removes the same bias with only a small change from the initial guess such that the time of RPSM does not shift much. And so, the process is iterated until the events occur within a second of the specified time. It takes a few iteration to achieve convergence with 1 second of accuracy.

Another interesting discovery with the initial guess using the mean period arithmetic is that by considering not only the “closer” point from the EDL time but also considering the “farther” point we were able to bring down the maximum RPSM ΔV from ~ 4 m/s to ~ 2.5 m/s. To elaborate on this more, for example, a given reference trajectory usually does not cross the specified latitude at the given EDL time. Therefore, the EDL latitude events must be searched for and the closest times when the events occur are taken for each EDL events. However, when we take the closest points at both ends, we unintentionally constrain the problem in a particular way that forces the solution to be non-optimal in some cases. Thus, when we consider both “closer” and “farther” (by farther, it refers to the same latitude crossing event but on the opposite side in time) possibilities at both EDL events, there are four solutions altogether. Often, a combination other than choosing the “closer” events yields a lower ΔV for RSMP.

To understand and verify this, an optimization problem was set up using a non-linear optimizer with two independent variables, (1) ΔP at APSM and (2) ΔP at RPSM and with two constraints, (1) the latitude crossing phase bias at InSight EDL and (2) the latitude crossing phase bias at EDM, both of which must be driven to zero. The cost of the optimization was to minimize the ΔV , which amounts to the same thing as minimizing the sum of the absolute value of the two ΔP s. Since the mean period propagation arithmetic worked so well with only a few percent error, we were actually able to solve the problem when a good set of initial ΔP guesses were given. Using this optimization problem, we not only found those four expected solutions but also many others. It turns out that when the APSM is farther away from the location where the phase is removed, its “characteristic” ΔP (the average period divided by the number of orbits between the maneuver and the time of phase removal) decreases to yield a solution that prefers the “farther” event. In fact, if the characteristic ΔP is lowered such that the number of the orbits decreases by one, we find another set of solutions, but usually with higher ΔV s. Thus, we verified that the ΔV at RSPM can be lower than the corresponding conventional phasing maneuver solution used in Phoenix and MSL EDL phasing (that cannot possibly be achieved between the two EDL events). Consequently, the ΔV was reduced from about 4 m/s to about 2.5 m/s.

C. Result Summary

Since the estimate method worked so well, with only a few percent difference from the fully optimized solution, this method was used exclusively to understand the scope of the problem. However, several spot checks were made to verify the validity of the estimates. For example, with the *phase+30°* trajectory and with the EDM event about 9/16 orbit out of phase from the InSight EDL, the fully optimized solution yielded 0.087 m/s for APSM (29-JUN-2016) and 2.320 m/s for RPSM (24-AUG-2016). Whereas the estimate method yielded -0.098 m/s for APSM (~ 0.01 m/s accuracy for a smaller burn) and 2.358 m/s for RPSM (about 2% error), which found the “farther” and the “farther” combination to be the most optimum. For another example, with the *phase0* trajectory and with the EDM event about 0.3 orbit out of sync from the InSight EDL, the fully optimized solution yielded 0.300 m/s for APSM (03-AUG-2016) and 0.335 m/s for RPSM (31-AUG-2016). Whereas the initial guess estimated 0.300 m/s for APSM and 0.335 m/s for RPSM (less than 1% error), which found the “closer” and the “closer” combination to be the most

* As already indicated, this is a trajectory phased by +30 minutes from *phase0*, which includes the phasing for the CSS from the MRO reference trajectory.

optimum. The second example had less error because the APSM and RPSM maneuver dates were closer to the EDL events—less error for less arithmetic mean period propagation. Note that the fully optimized solution takes on the order of hour whereas the estimate takes only a few seconds once the reference trajectory mean elements are obtained around the maneuver locations in the orbit.

We made many estimate runs, but the following sets of runs summarize a few points toward understanding the proposed maneuver strategy:

- 1) Given an RPSM date of 24-AUG-2016, an earlier APSM date yields a small savings for both RPSM ΔV and APSM ΔV — up to about 0.5 m/s in some cases. Please refer to Figure 8 below. Note that the X-axis represents a shifted EDM time up to 2 orbits. Essentially, the ΔV cost repeats every orbit as expected. Note also that the maximum RPSM is below 2.5 m/s and the maximum APSM is below 1 m/s.

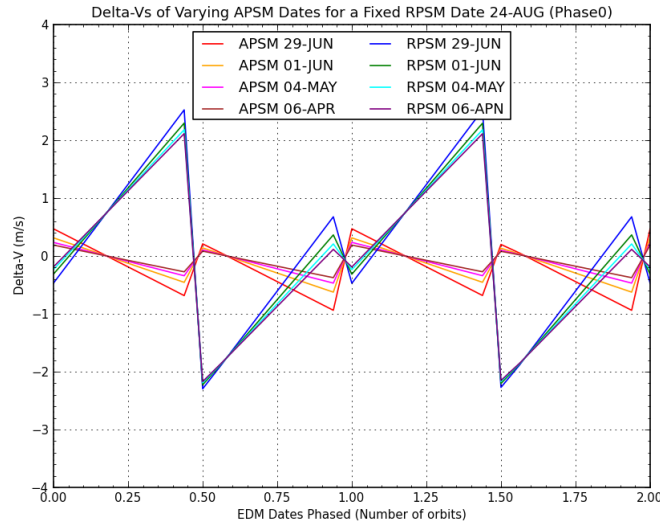


Figure 8. ΔV of Varying APSM Dates for a Fixed RPSM Date on 24-AUG-2016. *The reddish colors represent different APSM ΔV values at different APSM times, and the bluish colors represent different RPSM ΔV values at different RPSM times.*

- 2) Given an APSM date of 06-APR-2014, a later RPSM date yields a lower ΔV . This seems a bit odd at first, but the strategy works better when there is more time between APSM and RPSM. Refer to Figure 9 below for more detail.

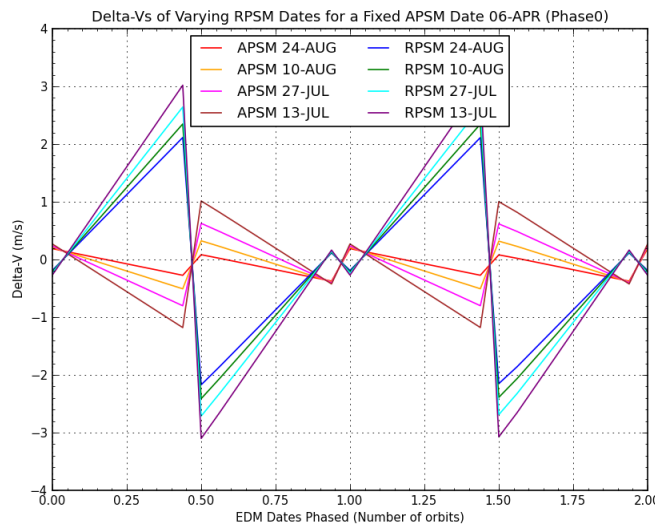


Figure 9. ΔV s of Varying RPSM Dates for a Fixed APSM Date on 01-APR-2016.

- 3) Trajectories with different phasing only shift the ΔV curve. It does not change the ΔV values themselves radically. Refer to Figure 10 below for more detail.

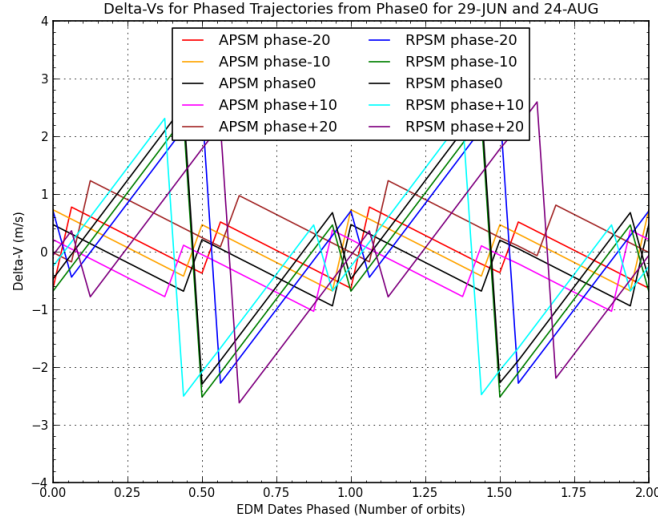


Figure 10. ΔV s of Varying Phase of the Reference Trajectory.

The above findings will be useful for arranging the maneuver locations in operations as well as for preparing for anomalies with confidence.

D. Robustness

The robustness of the strategy is shown by a sequence of downtrack timing uncertainty propagations. By inputting sufficiently conservative estimates into the downtrack timing uncertainty propagation and carrying through those assumptions from the start of APSM to the end of EDM, we show that the final timing uncertainties at InSight EDL and at EDM are below the level of the requirements. Due to page limitations, we show only one case. The following inputs include conservative estimates for the downtrack timing uncertainty propagation:

- 1) Use the latest available date for APSM, 29-JUN-2016. Earlier dates would yield a slightly lower ΔV for APSM.
- 2) Use the last available date for RPSM, 24-AUG-2016.
- 3) Perform a final EDM target maneuver on 08-OCT-2016, which is 10 days after InSight EDL and is sufficiently far away from InSight EDL operationally.
- 4) Use higher ΔV values for APSM and RPSM, 2 m/s and 4 m/s, respectively (although we have shown that the maximum ΔV can be lowered to 1 m/s and 2.5 m/s).
- 5) Use very conservative atmospheric drag values of 0.5 mm/s per orbit between APSM (29-JUN-2016, 177.1° Ls) and RPSM (24-AUG-2016, 209.8° Ls) and 0.7 mm/s per orbit after RPSM.
- 6) The orbit determination data cut-off (DCO) for maneuvers is set to 7 days, a nominal standard for all other leisurely MRO maneuvers.

Figure 11 below shows the timeline diagram of the events. Two additional cleanup maneuvers are inserted for APSM and RPSM: (1) APSM cleanup 2 weeks prior to RPSM, on 10-AUG-2016; and (2) RPSM cleanup 2 weeks before InSight EDL, on 14-SEP-2016.

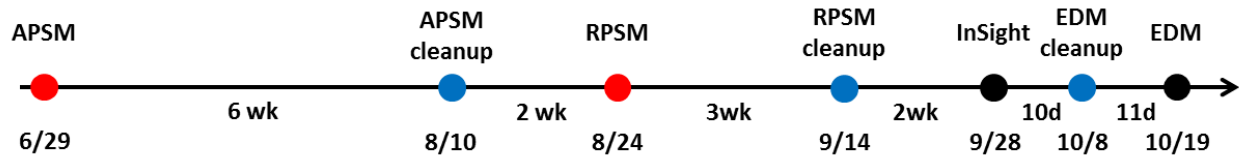


Figure 11. Timeline for Downtrack Timing Uncertainty Propagation. The deterministic APSM & RPSM maneuvers are in red. The statistical cleanup maneuvers are in blue. The NSYT and EDM events are in black. The dates MM/DD is in 2016. The events are marked by the duration in either weeks or days.

Figures 12-16 below show five plots as a sequence of downtrack timing propagation from APSM to EDM. They are divided from one maneuver to the next: (1) from APSM to APSM cleanup; (2) from APSM cleanup to RPSM; (3) from RPSM to RPSM cleanup; (4) from RPSM cleanup to EDM cleanup; and (5) from EDM cleanup to EDM. The resulting RSS value of the timing uncertainty is carried on as an additional ΔV into the following case.

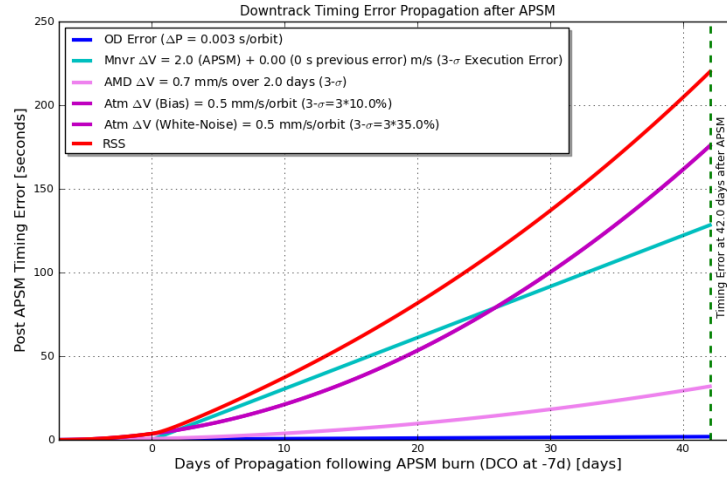


Figure 12. Timing Uncertainty from APSM to APSM Cleanup.

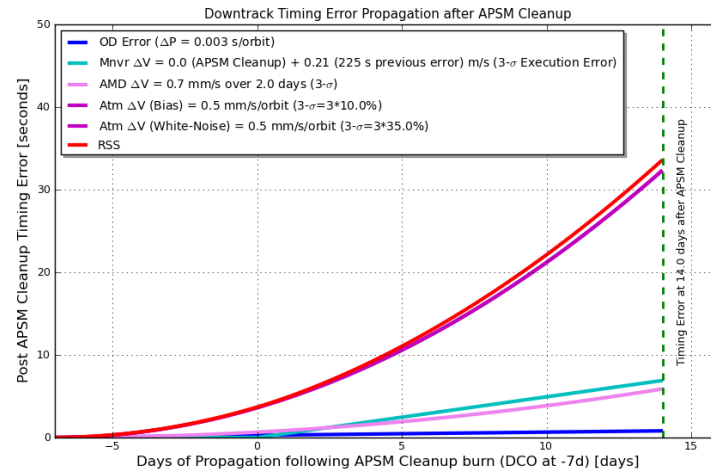


Figure 13. Timing Uncertainty from APSM Cleanup to RPSM.

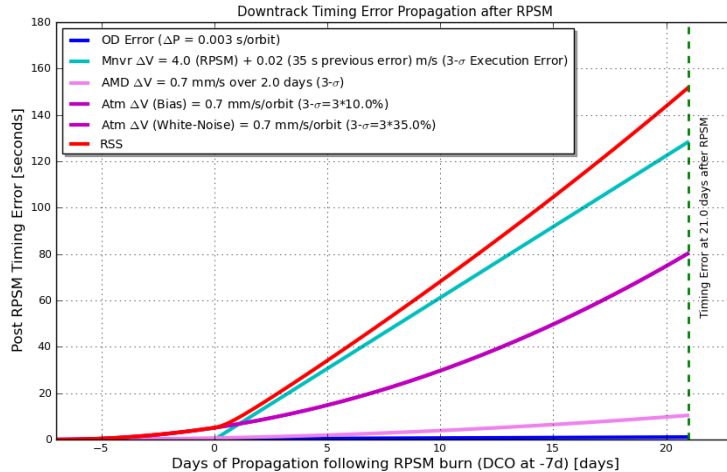


Figure 14. Timing Uncertainty from RPSM to RPSM Cleanup.

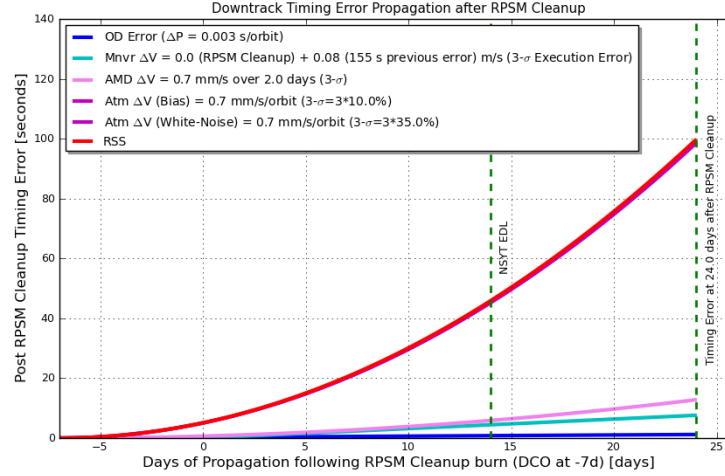


Figure 15. Timing Uncertainty from RPSM Cleanup to EDM Cleanup.

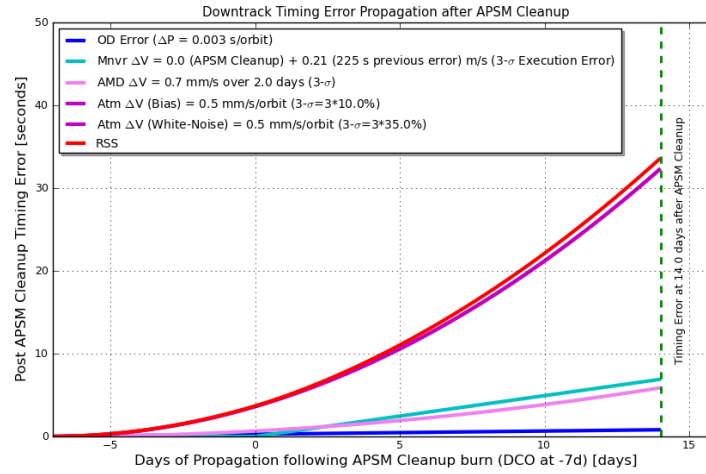


Figure 16. Timing Uncertainty from EDM Cleanup to EDM.

Even with the most conservative assumptions the proposed strategy meets the phasing requirements. In reality, the conservative ΔV and the atmospheric drag estimates used above may be relaxed to yield a better result than shown here.

IV. Conclusion

Although the proposed maneuver strategy appears somewhat radical at first, having never been tried, it is shown in this paper that the proposed maneuvers should behave similar to any other phasing maneuvers used previously by MRO. The ability to design such a maneuver set to solve the problem has been demonstrated. A better understanding of the proposed strategy's behavior and its robustness should help encourage its use in the flight operations environment.

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References

- ¹Edwards, C., Bruvold, K., Erickson, J., Gladden, R., Guinn, J., Ilott, P., Jai, B., Johnston, M., Kornfeld, R., Martin-Mur, T., McSmith, G., Thomas, R., Varghese, P., Signori, G., and Schmitz, P., "Telecommunications Relay Support of the Mars Phoenix Lander Mission," *Aerospace Conference*, 2010 IEEE, vol., no., pp. 1-13, March 6-13, 2010.

²Williams, J., Menon, P., and Demcak, S., “Mars Reconnaissance Orbiter Navigation Strategy for Mars Science Laboratory Entry, Descent and Landing Telecommunication Relay Support,” AIAA/AAS Astrodynamics Specialist Conference, AIAA 2012-4747, Minneapolis, Minnesota, August 13-16, 2012.

³Konopliv, A., “A Third Order J2 Solution with a Transformed Time,” Interoffice Memorandum, Jet Propulsion Laboratory, 314.3-970, 28 March, 1991.

⁴Kaula, W., *Theory of Satellite Geodesy*, Dover Publications, Inc., Mineola, New York, 2000.